ME530

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Mass Impacting Solid Barrier

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1. Introduction

This report presents the modeling and simulation of a mechanical system in the software 20-Sim. The system has one dimension of motion, and 4 independent energy storing elements (3 springs, 1 mass). A spring is attached to the mass and has a sinusoidal velocity input matching the resonant frequency of the spring-mass, which causes increasing oscillations until the spring-damper barriers are impacted. The effect of input frequency, barrier stiffness, and spring-mass resonant frequency on barrier penetration depth and overall response are analyzed See figures below.

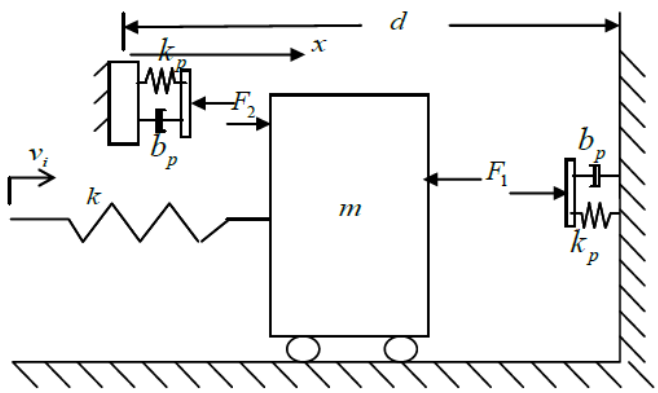

Figure 1: Physical Model

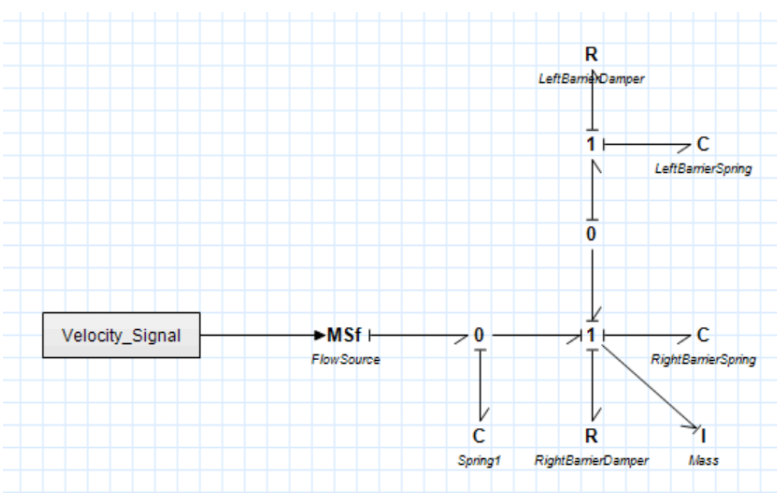

Figure 2: Bond Graph Model

2. Analysis

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The spring rate k is calculated to cause a 1Hz resonant frequency for a mass of 1kg.

$$f_{natural} = 1Hz$$

$$k = 1kg * (2\pi f_{natural})^2 = 39.4784N/m$$

A global variable is made in 20-Sim, called displacement, which is accessed by the C-elements (barrier springs) and R elements (barrier dampers), and is the integral of the flow (velocity). When the mass displacement exceeds $d = 5m$ in the positive direction, the right barrier C and R elements are activated. When the mass displacement becomes negative, the left barrier C and R elements are activated. The initial displacement value is 0.25m, when the mass is unmoving and the attached spring is uncompressed. The velocity_signal block is shown below.

```
equations
   y =mag*sin(2*3.14*freq*time);
```

Figure 3: Velocity Signal Block

Where:

$$mag = x_i * (2\pi f_{natural}) = 0.25m * (2\pi * 1Hz) = 1.57$$

$$freq = f_{natural} = 1Hz$$

For tests 1-3 left and right barriers were tested with stiffnesses 100, 1000, and 10000 times greater than the stiffness of the mass-attached spring. Corresponding barrier damping was chosen for a damping ratio $\zeta = 0.05$. For tests 4-6, different damping ratios were chosen for the barrier. In tests 7-9, forcing frequency is changed to half, twice, and three times the natural frequency of the spring-mass system. Test 10 uses a spring-mass system with a higher natural frequency, and matching forcing frequency. See the test tables below.

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			Test Number		
			1 ($\frac{k_p}{k_1} = 100$)	**2** ($\frac{k_p}{k_1} = 1000$)	**3** ($\frac{k_p}{k_1} = 10000$)
System	Parameter/ Element	Bond Element	Value		
Spring-Mass	Mass	I	$1kg$		
	Spring1	C	$39.4784N/m$		
	$f_{natural}$	-	$1Hz$		
	Input Flow	MS_f	$v(t) = 1.57 * cos(2\pi t)$		
Right Barrier	Spring2	C	$3947.84N/m$	$39478.4N/m$	$394784N/m$
	Damper2	R	$6.2832\ Ns/m$	$19.8692\ Ns/m$	$62.8318\ Ns/m$
	ζ	-	0.05		
Left Barrier	Spring3	C	$3947.84N/m$	$39478.4N/m$	$394784N/m$
	Damper3	R	$6.2832\ Ns/m$	$19.8692\ Ns/m$	$62.8318\ Ns/m$
	ζ	-	0.05		

Figure 4: Testing of different barrier stiffnesses

			Test Number		
			4 ($f_i = 0.5f_{natural}$)	**5** ($f_i = f_{natural}$)	**6** ($f_i = 2f_{natural}$)
System	Parameter/ Element	Bond Element	Value		
Spring-Mass	Mass	I	$1kg$		
	Spring1	C	$39.4784N/m$		

	$f_{natural}$	-	$1Hz$		
	Input Flow	MS_e	$v(t) = 1.57 * cos(2\pi * 0.5 * t)$	$v(t) = 1.57 * cos(2\pi * 1 * t)$	$v(t) = 1.57 * cos(2\pi * 2 * t)$
Right Barrier	Spring2	C	$39478.4N/m$		
	Damper2	R	$19.8692\ Ns/m$		
	ζ	-	0.05		
Left Barrier	Spring3	C	$39478.4N/m$		
	Damper3	R	$19.8692\ Ns/m$		
	ζ	-	0.05		

Figure 5: Testing of different forcing frequencies

			Test Number		
			7	**8**	**9**
System	Parameter/ Element	Bond Element	Value		
Spring-Mass	Mass	I	$1kg$		
	Spring1	C	$39.4784N/m$		
	$f_{natural}$	-	$1Hz$		
	Input Flow	MS_e	$v(t) = 1.57 * cos(2\pi t)$		
Right Barrier	Spring2	C	$39478.4N/m$		
	Damper2	R	$19.8692\ Ns/m$	$80\ Ns/m$	$240\ Ns/m$
	ζ	-	0.05	~ 0.2	~ 0.6
Left Barrier	Spring3	C	$39478.4N/m$		
	Damper3	R	$19.8692\ Ns/m$	$80\ Ns/m$	$240\ Ns/m$
	ζ	-	0.05	~ 0.2	~ 0.6

Figure 6: Testing of different barrier damping ratios

			Test Number
			10
System	Parameter/ Element	Bond Element	Value
Spring-Mass	Mass	I	$1kg$
	Spring1	C	$3947.84N/m$
	$f_{natural}$	-	$10Hz$
	Input Flow	MS_e	$v(t) = 1.57 * cos(10 * 2\pi t)$
Right Barrier	Spring2	C	$39478.4N/m$
	Damper2	R	$19.8692\ Ns/m$
	ζ	-	0.05
Left Barrier	Spring3	C	$39478.4N/m$
	Damper3	R	$19.8692\ Ns/m$
	ζ	-	0.05

Figure 6: Test 10 uses same values as Test 2, but increased Spring1 stiffness for a spring-mass natural frequency of 10Hz, and matching 10Hz driving frequency

3. Results

With barriers removed, the system is unstable as expected and resonates at 1Hz.

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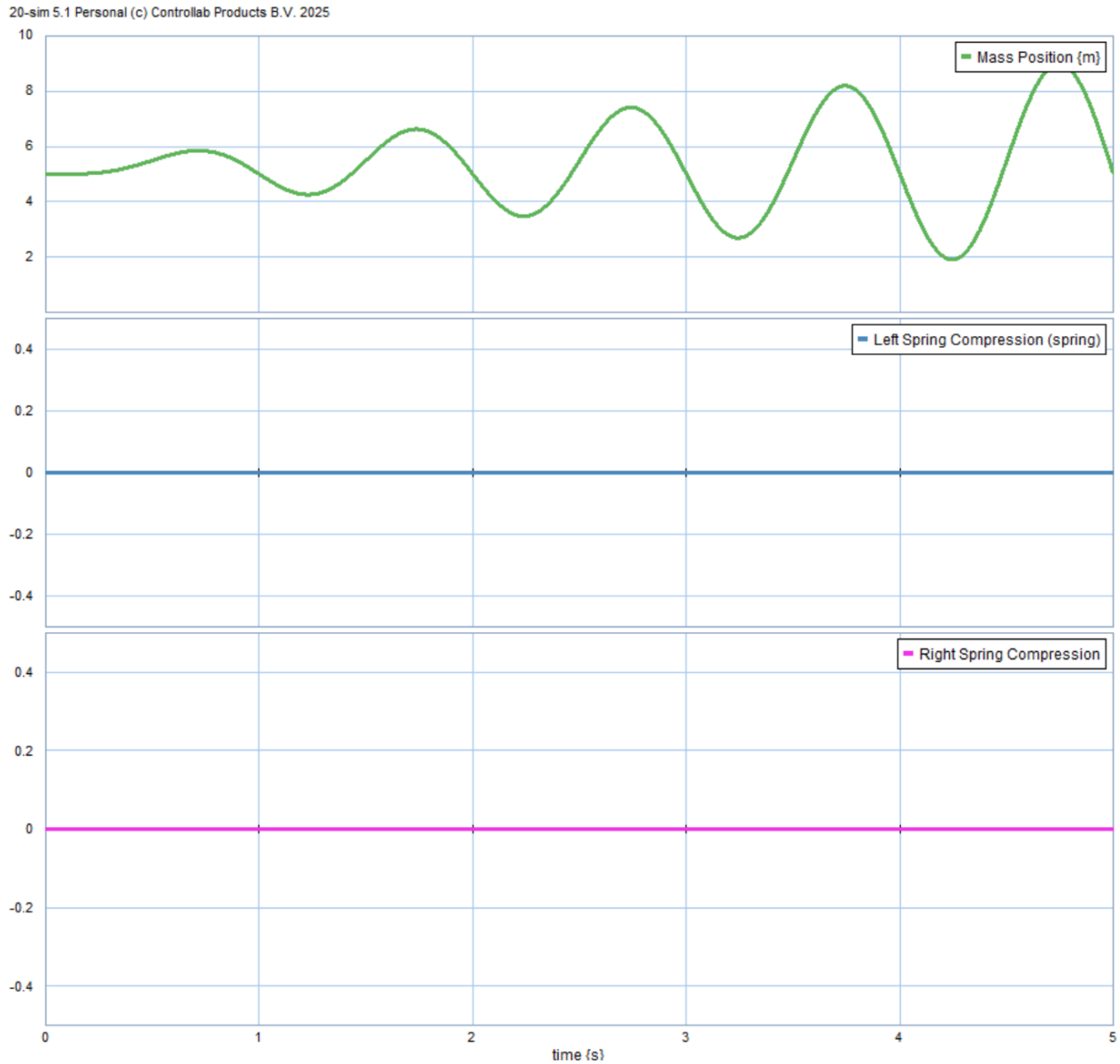

Figure 7: No barriers, system resonates at 1Hz

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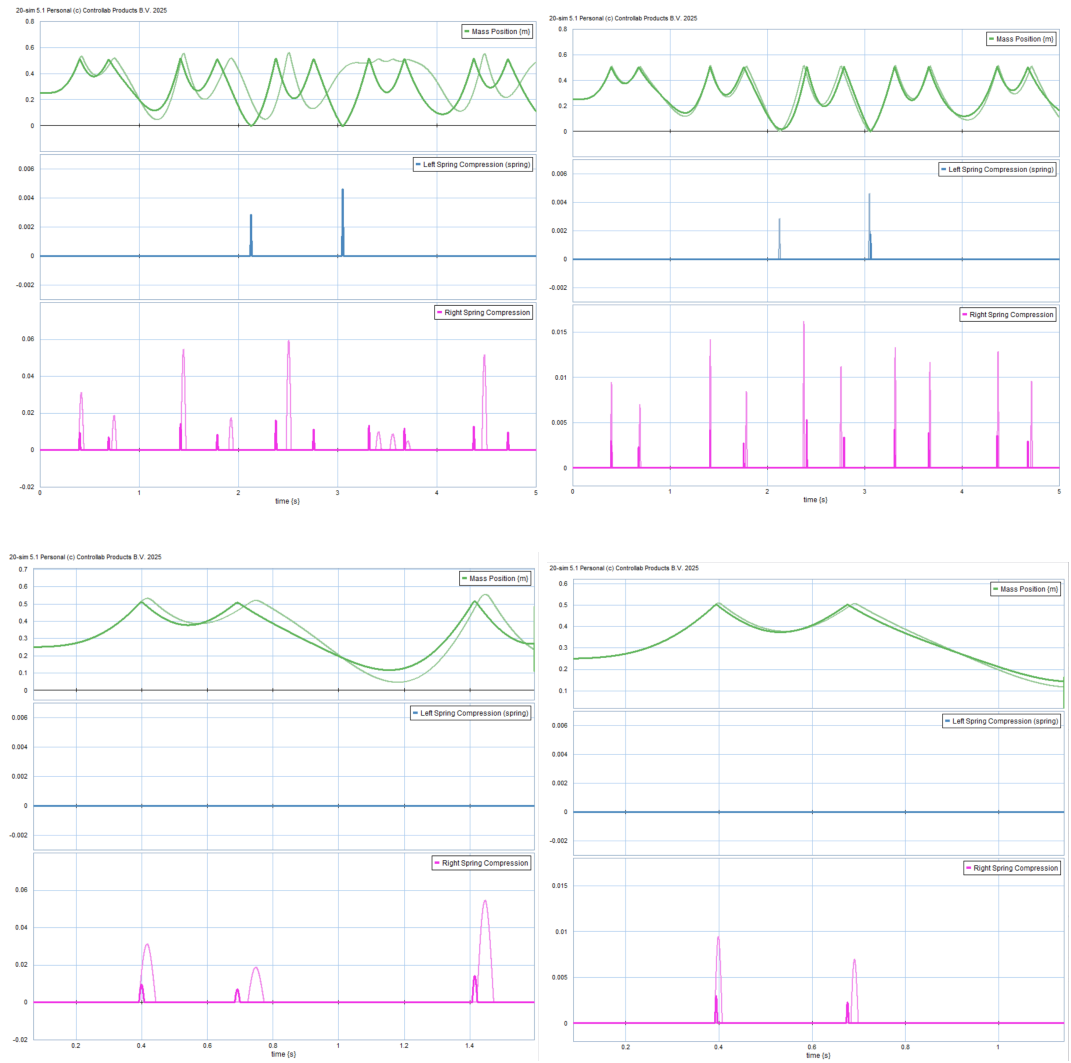

Figure 8: Effect of barrier stiffness on maximum penetration. At left: $100 * k_1$ (Test 1, light) to $1000 * k_1$ (Test 2, dark) reduces barrier penetration from ~31mm to ~9mm. At right: $1000 * k_1$(Test 2, light) to $10000 * k_1 1000 * k_1$(Test 3, dark) reduces barrier penetration from ~9mm to ~3mm.

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Figure 9: Effect of drive frequency. At left: Test 4 (light) with 0.5Hz drive frequency and Test 5 (dark) with 1Hz drive frequency. At right: Test 5 (light) with 1Hz drive frequency, and Test 6 (dark) with 2Hz drive frequency.

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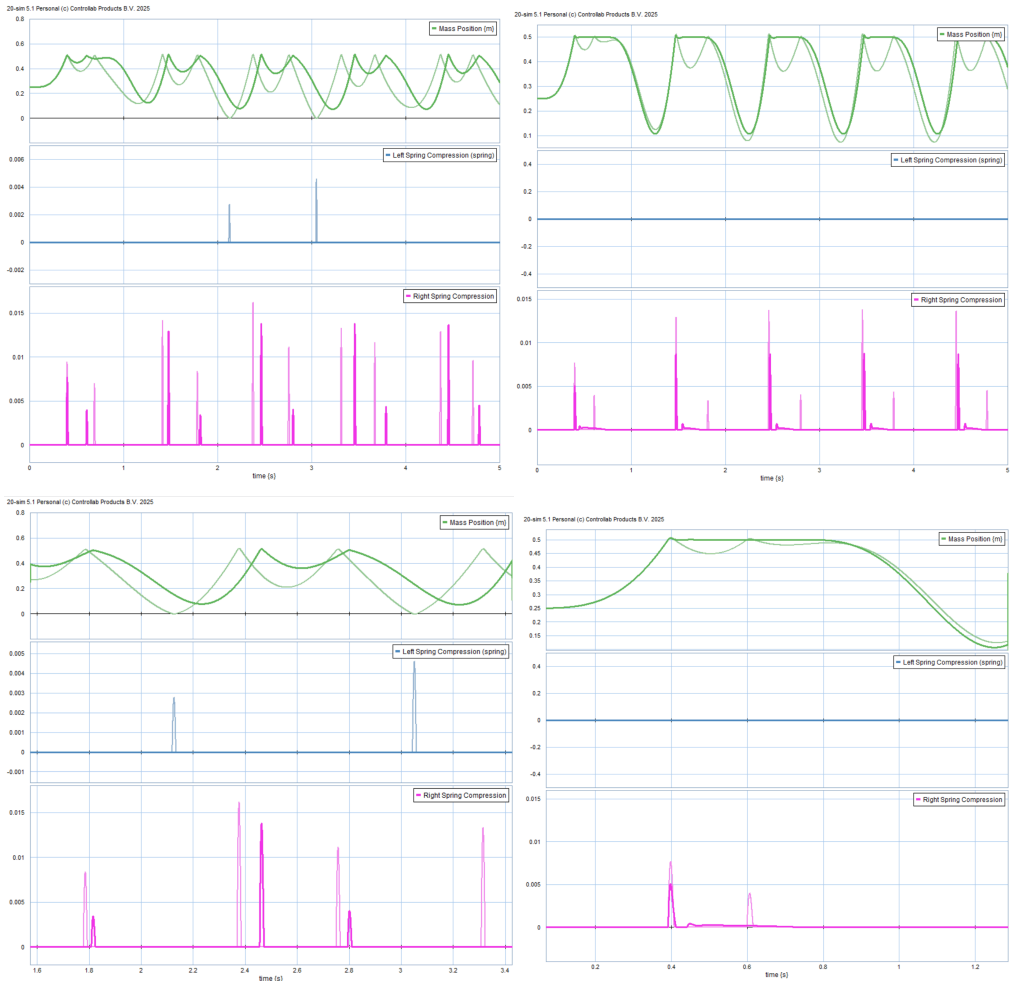

Figure 10: At left, increasing damping ratio from $\zeta = 0.05$ (Test 7, light) to $\zeta = 0.2$ (Test 8, dark) slightly reduced the right barrier penetration, and eliminated 2 instances of light contact with the left barrier. At right, increasing the damping ratio from $\zeta = 0.2$ (Test 7, light) to $\zeta = 0.8$ caused the mass impacts with the barrier to no longer ‘bounce’, but instead remain in contact with the right side barrier for ~200ms each cycle.

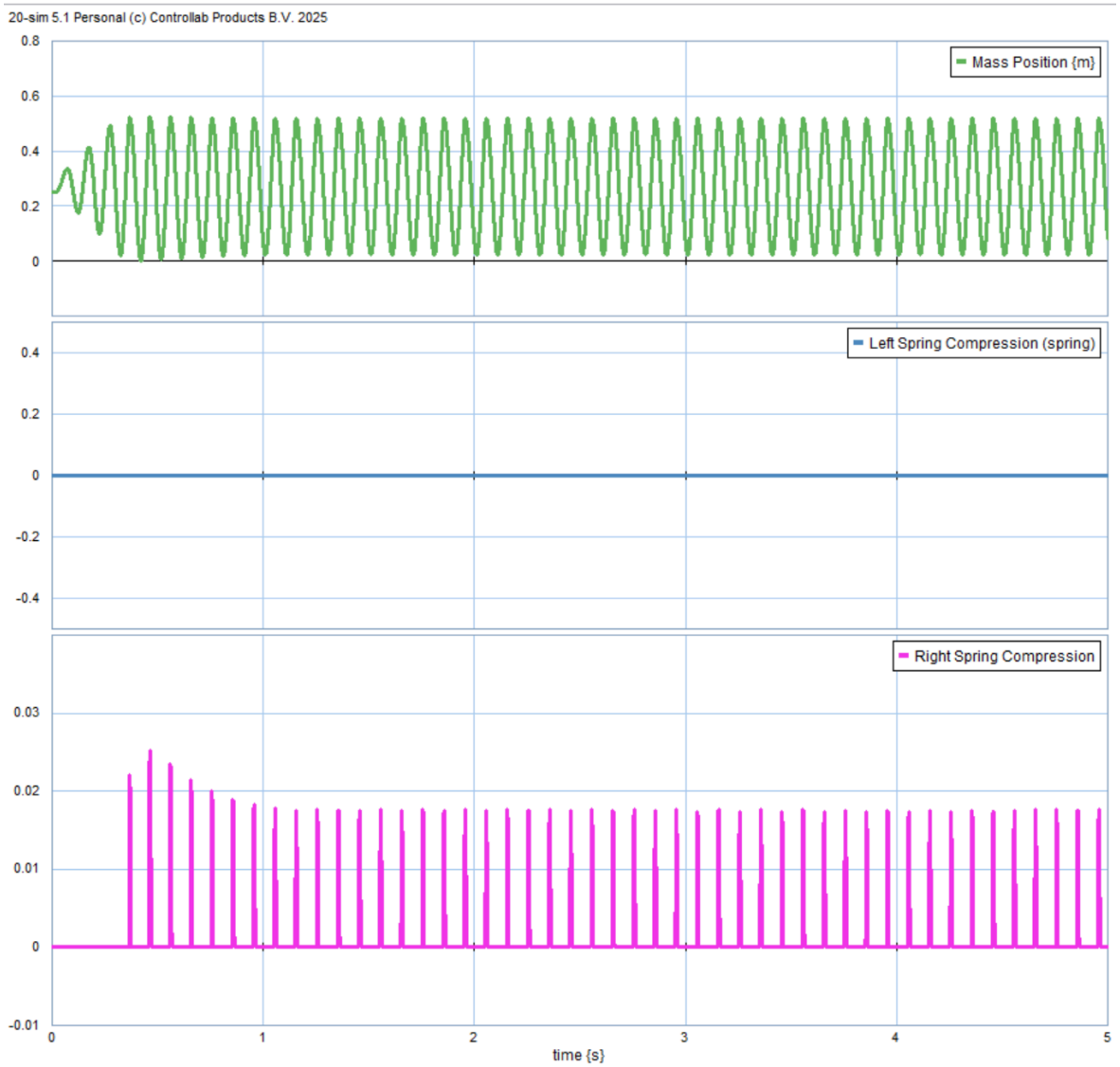

Figure 11: System response with 10Hz natural frequency and 10Hz driving frequency

4. Discussion

In tests 1-3 the mass oscillated at roughly 1Hz while bouncing off of the right barrier twice per cycle. In test 1, a barrier stiffness of $k_{barrier} = 100k_1$ resulted in ~31mm of penetration on the right barrier, and did not contact the left barrier. Increasing the stiffness 10x (Test 2) reduced the penetration to ~9mm on the right barrier, while 100x (Test 3) resulted in the lowest barrier penetration of ~3mm. After the first contact with the right barrier during each cycle, the mass rebounded 100-250mm before hitting the right barrier again. This is likely due to the relatively low damping ratio ($\zeta = 0.05$), where very little energy is absorbed by the barrier dampers and instead stored in spring potential and then returned to the mass causing a “bounce”. This reasoning is supported by tests 8-9 where increasing the damping ratio from

0.05 to 0.2 causes this rebound to shrink significantly (see figure 10, top left plot). Test 9-10 increases the damping ratio further from 0.2 to 0.8, and causes the mass to “stick” to the right side barrier each cycle, as opposed to the bouncing behavior with lower damping ratios. Higher damping ratios are therefore better if the goal is to absorb energy from the mass and reduce chattering. Reducing the drive frequency to 0.5Hz causes the mass to bounce off of the right side barrier repeatedly before swinging back in the opposite direction, where each subsequent bounce decreases in magnitude as the damper slowly dissipates the total energy in the system. A higher (2Hz) drive frequency causes single-contacts with the barriers. In the case of the system with 10Hz natural frequency and matching 10Hz drive frequency, single contacts (no bouncing) against the right barrier took place, although more simulations should be run for this high natural frequency system to draw strong conclusions.

5. Conclusion

While increasing the damping ratio helped to reduce barrier penetration, it mostly served to reduce bouncing and chattering of the mass against the barrier while swinging back and forth at the drive/resonant frequency. To minimize penetration, the stiffness of the barrier should be tuned, where in our simulations a 10-fold increase in barrier stiffness reduced the barrier penetration by ⅓.